AI FAQ & Customer Support Automation

PROJECT PLANNING & EXECUTION ROADMAP

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Project Domain: AI-powered customer support backend & automated FAQ resolution

PHASE PLAN

Phase 1: Identify problem, target users, support pain points, and AI concept.

Phase 2: Define functional/non-functional, data, security, software, and hardware requirements.

Phase 3: Design architecture, ER/data model, APIs, security, and AI flow.

Phase 4: Plan modules, milestones, testing strategies, and submission criteria.

Phase 5: Implement Node.js/Express, MongoDB/Mongoose, JWT, ticket/FAQ CRUD/search, and Gemini integration.

Phase 6: Test APIs, validation, authentication, CRUD, search, and AI endpoints.

Phase 7: Finalize technical documentation, API specs, and presentation deck.

Phase 8: Record and prepare the live project demonstration.

Work Breakdown

- Authentication & Role-based Security
- FAQ & Knowledge Base CRUD / Search
- Support Ticket Lifecycle & Escalation
- Gemini Query Resolution & Auto-reply
- Input Validation & Error Handling
- API Testing & QA
- Documentation, Presentation & Demo

Milestones Workflow

Concept → Requirements → Core Backend Design → FAQ & Support Module → AI Integration → Testing → Documentation / Demo

Sequential execution ensuring backend robustness prior to AI layer integration and QA.

COMPLETION CRITERIA

- Backend server runs successfully and connects to MongoDB without errors.
- Protected routes require and correctly validate JWT authentication.
- FAQ and Support Ticket CRUD/search operations execute reliably.
- Gemini AI endpoints process queries and output relevant responses/auto-replies.
- API test suites pass and are fully documented.
- Final project documentation, presentation slides, and demonstration video are prepared.